



ASTRIX — ISLAND VILLAGE

3D MODELING SHEET / GAME-READY ASSET BIBLE • Blender → glTF/GLB → Three.js • v1.0 Rebuild

PRIMARY: Image 3 = Final Visual Target

SUPPORT: Image 1 = Dark Tech Sheet

SUPPORT: Image 2 = Light Asset Sheet

Style: Stylized Low/Mid-Poly Tropical

Platform: Web Three.js 60 FPS

Format: GLB (glTF 2.0)

A LIVING WORLD.
BUILT FOR REAL-TIME.

Same world. Many stories.

REFERENCE PRIORITY — READ FIRST. Image 3 (hero isometric) is the PRIMARY visual target for the final Three.js world: dense tropical settlement, town-hall plaza with monument fountain, farms upper-right, windmill on ridge, lighthouse on SE rock, wooden docks/pier, waterfalls, beaches, coral reefs, turquoise water. **Images 1 + 2** are supporting vocabulary for asset construction: modular buildings, terrain modules, vegetation silhouettes, props, NPCs, boats, materials, ortho views, wireframe/construction language, Blender notes. **Do NOT build one giant mesh.** Build modular, instanced, game-ready assets and assemble the island from them.

1 — MASTER WORLD REFERENCE

Hero view + ortho views + terrain zones. All modeling must match this look while staying within browser budgets. Target: stylized low-poly/mid-poly, painterly, readable silhouettes, warm sun, turquoise water, no photorealism, no primitive-only placeholders.

HERO — FINAL SCENE REFERENCE (Image 3 — PRIMARY)





Island Size ~800×600 m (world units scaled to Three.js: 1 unit = 1 m)

Main Island ~500×350 m

Water Depth 5–40 m (visual only, no sim)

Buildings 45+

Trees/Plants 200+

Props 300+

Camera Isometric / elevated 45° yaw, ~38° pitch, ortho or low-FOV persp

ORTHO — TOP / FRONT / SIDE (from Images 1+2)

TECH MODELING SHEET (Image 1 — SUPPORT)



ASTRIX ISLAND VILLAGE 3D MODELING SHEET



FINAL SCENE REFERENCE
(Target Visual Style)

SCENE OVERVIEW



Top View



Perspective View



Side View

SCENE SCALE (approx.)

Island Size	800 x 600 m
Main Island	~500 x 350 m
Water Depth	5 - 40 m
Buildings	45+
Trees/Plants	200+
Props	300+

BUILDING ASSETS (Modular & Stylized)

Residential House

3.2m x 3.5m x 4.0m
Tris: 2,842

Town Hall

5.8m x 5.2m x 6.5m
Tris: 5,620

Market Stall

3.5m x 2.8m x 3.0m
Tris: 1,240

Farm House

4.2m x 4.0m x 4.5m
Tris: 2,310

Windmill

5.0m x 5.0m x 7.0m
Tris: 3,890

Lighthouse

4.0m x 4.0m x 8.0m
Tris: 2,760

Greenhouse

4.5m x 3.5m x 3.5m
Tris: 2,180

Dock / Pier

8m x 3m x 2.5m
Tris: 1,120

Bridge

6m x 4m x 2.5m
Tris: 1,480

Temple / Monument

4m x 2.5m x 2m
Tris: 820

Storage Barn

2m x 2m x 0.2m
Tris: 320

Mine / Cave Entrance

2m x 2m x 0.2m
Tris: 320

PROPS & DETAILS



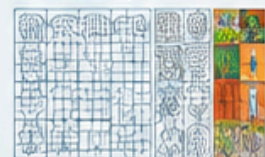
VEGETATION & ENVIRONMENT



MATERIALS & TEXTURES



UV LAYOUT (example)



TEXTURE ATLAS (example)



CHARACTERS (NPCs)



BOATS



BLENDER SETUP NOTES

- Use Blender 4.x
- PBR materials (Principled BSDF)
- UV unwrapping for game-ready assets
- Target polygon budget: 1k - 10k tris (per asset)
- Use consistent scale (1 Blender unit = 1 meter)
- Export as glTF / GLB for Three.js
- Keep texture resolution optimized (1024-2048)
- Use vertex colors or simple shaders for variation



Modeled for a living world
Built for Three.js



ASTRIX ISLAND VILLAGE 3D MODELING SHEET / GAME-READY ASSETS

A LIVING WORLD.
BUILT FOR REAL-TIME.



TARGET VISUAL

Stylized • Low Poly • Tropical • Isometric

WORLD LAYOUT



- 1 Main Island (Settlement)
- 2 Farm Zone
- 3 Windmill
- 4 Lighthouse
- 5 Dock / Pier
- 6 Bridge
- 7 Waterfall
- 8 Smaller Island
- 9 Coral Reef

SCENE OVERVIEW



Island Size (approx.)	800 x 600 m
Main Island	~500 x 350 m
Water Depth	5 - 40 m
Buildings	45+
Trees/Plants	200+
Props	300+

ORTHOGRAPHIC VIEWS (Main Island)



TECHNICAL SPECS

Target Platform	Web (Three.js)
Render Style	Stylized / PBR
Terrain Tiles	256x256 grid
Model Format	GLB (glTF 2.0)
Texture Size	1024 / 2048 (px)
Max Draw Calls	< 200
LOD	3 levels
Frame Target	60 FPS

COLOR PALETTE



BUILDINGS (Game Ready - Modular)

Residential House

3.2m x 3.5m x 4.0m
Tris: 2,842

Town Hall

5.8m x 5.2m x 6.5m
Tris: 5,620

Market Stall

3.5m x 2.8m x 3.0m
Tris: 1,240

Farm House

4.2m x 4.0m x 4.5m
Tris: 2,310

Windmill

5.0m x 5.0m x 7.0m
Tris: 3,890

Lighthouse

4.0m x 4.0m x 8.0m
Tris: 2,760

Greenhouse

4.5m x 3.5m x 3.5m
Tris: 2,180

TERRAIN & ENVIRONMENT

Island Cliff

8m x 6m x 6m
Tris: 1,820 (LOD)

Beach Edge

10m x 6m x 2m
Tris: 4,000 (LOD)

Waterfall

6m x 8m x 10m
Tris: 2,140 (LOD)

INFRASTRUCTURE

Bridge

8m x 3m x 2.5m
Tris: 1,120

Dock

6m x 4m x 2.5m
Tris: 1,480

Pier

4m x 2.5m x 2m
Tris: 820

Path (Segment)

2m x 2m x 0.2m
Tris: 320

VEGETATION

Palm Tree

4m x 4m x 6m
Tris: 420

Tropical Tree

5m x 5m x 7m
Tris: 680

Bush

2m x 2m x 2m
Tris: 260

Bamboo

1.5m x 1.5m x 4m
Tris: 180

Flower Cluster

1.5m x 1.5m x 1m
Tris: 120

PROPS & DECOR

Crate

1m x 1m x 1m
Tris: 120

Barrel

0.8m x 0.8m x 1m
Tris: 80

Market Stall

3.0m x 2.0m x 2.5m
Tris: 980 (LOD)

Sign Post

1.5m x 0.3m x 2.5m
Tris: 70 (LOD)

Lantern

0.4m x 0.4m x 1.2m
Tris: 60

VEHICLES

Fishing Boat

5.5m x 2.5m x 2.5m
Tris: 1,160

Sailboat

6.5m x 3.0m x 5.0m
Tris: 1,420

Rowboat

3.5m x 1.5m x 0.8m
Tris: 720

CHARACTERS (NPCs)

Villager (Male)

Tris: 1,120

Villager (Female)

Tris: 1,080

Farmer

Tris: 1,240

Fisherman

Tris: 1,180

Guard

Tris: 1,320

TEXTURES & MATERIALS (Examples)

Albedo

1024x1024

Normal

1024x1024

Roughness

1024x1024

ORM

1024x1024

LODs (Example: House)

LOD 0

(2,842 tris)

LOD 1

(1,240 tris)

LOD 2

(520 tris)

Modular • Optimized • Game Ready • Three.js

Same world. Many stories.



Follow its construction language: dimension lines, wireframe cage next to shaded view, tris + meters under each asset, material dots, LOD strip, texture swatches. Reproduce this layout discipline for every new asset.

Use Image 2 top-view for plaza→farm axis, side-view for cliff/waterfall heights, perspective for density.
Use Image 1 ortho strip for topology validation.

Key Terrain Zones (label for placement — see §2)

Zone	Location on hero	Modeling note
Z1 Central settlement + plaza	Middle, around fountain monument	Flattened pad ~60×50 m, stone plaza disc, radial paths. Keep clear of scatter.
Z2 Residential ring	Around plaza, W + S	8–12 houses on 8–12 m lots, varied roof color/rotation, no identical neighbors.
Z3 Market row	E of plaza	Stalls + storage barn, crates/barrels, banners, lanterns.
Z4 Farm belt	NE plateau	4–6 tilled plots, farmhouses, greenhouse, windmill on ridge behind.
Z5 Cliffs + waterfalls	N ridge + W arms	Modular cliff kit, 3 waterfalls (N big, W double, SW small), foam + mist planes.
Z6 Beaches + shallows	S + SE + E coves	Sand overlap onto grass, no floating discs; wet-rock band at waterline.
Z7 Harbor: dock/pier + lighthouse rock	SE	L-dock + lighthouse islet + bridge; sailboat + rowboat moored.
Z8 Outer islets + reefs	SW hut islet, S reef, E rocks, N skerries	Small rocky islands (same kit, scaled), coral + reef rocks + seaweed underwater.

2 — WORLD LAYOUT (approximate placement map)

Coordinates in meters, origin at plaza fountain (0,0). +X east, +Z south. Keep plaza→farm axis readable from default camera (upper-left key light).

Placement Table (tolerance ±10 m — compose, don't grid-snap)

Element	Approx. pos	Footprint
Central plaza + monument fountain	0,0	Ø 18 m stone disc

Composition Rules

- 1 landmark per view: plaza spire from S, windmill from N, lighthouse from E. Never cluster all three.
- Foreground corridor: keep S slope (camera side) low — bushes/tufts/rocks only, no tall trees blocking plaza.
- Tall vegetation: interiors + N/W rims. Palms on beaches, broadleaf inland, bamboo near water.

Town Hall (blue roof)	0,-28 (N of plaza)	9×8 m	• Water reads as water: boats in frame, foam at shore (shader or geo band), cliffs continue 5 m below waterline, seabed plane at -6 m, wet dark band at waterline.
Residential W cluster (4–5 houses)	-45...-20, -10...+25	lots 10 m	• Food story forward: farms face default camera; granary/stacks near market, visible gold accent.
Residential S cluster (3–4 houses)	-15...+20, +28...+50	lots 10 m	• Noxisolation: every building gets path + 1–2 props + 1 lantern + vegetation neighbor.
Market row + Storage Barn	+35,+5	30×14 m strip	Path Network
Farm belt (4–6 plots) + 2 Farm Houses	+70...+130, -60...-10	plots 12×8 m each	Path = 2 m wide segments (straight / curve 45° / T / plaza ring). Y-offset +0.06 m thin boxes (never coplanar quads). Sandy #E8CF9A with stone edging. Stairs where slope > 20°.
Greenhouse	+135,-15	7×5 m	MODULAR ASSEMBLY: terrain = tiled base (16–32 m tiles) + cliff kit + beach kit + islet kit. Never one monolithic island mesh — enables frustum culling + LOD + reuse.
Windmill (on ridge)	+55,-85	7×7 m, blades Ø 12 m	
Water tower / lookout	+110,-55	Ø 4 m	
Main dock + harbor house (SE)	+75,+95	dock 30×10 m	
Lighthouse rock	+125,+80	islet Ø 22 m, tower h 12 m	
Footbridge to lighthouse	+100,+88	span ~24 m	
SW fishing hut islet + pier	-60,+110	islet Ø 26 m	
S reef / coral gardens	-10...+40,+130	60×25 m shallow	
N waterfall cliff	-10,-110	cliff 40 w × 25 h m	
W twin falls + beach cove	-110,+20	beach 40 m arc	
Paths	plaza radial N/E/S/W + farm lane + harbor descent	2 m wide, thin-box decals	

3 — BUILDING ASSET SHEET (8 modular assets)

Each asset: shaded front + side + ¾ + wireframe cage. Origin at ground center. Units meters. All quads→tris, apply transforms before export. Shared atlas (\$12).

A. Residential House

Size 5.0×4.2×4.5 m **Tris LOD0** 2,842 **Tex** 1024 atlas

- **Modules:** base walls, gable roof (2 slabs + ridge cap + eaves), chimney + smoke anchor, door + 2 windows + shutters, porch posts, foundation skirt.
- **Materials:** plaster wall, roofTile orange/blue/red variants, timber trim, glass.
- **Variants:** 3 roof colors, mirror plan, porch on/off → 6 SKUs from 1 kit.
- **LOD:** LOD1 1,340 (drop shutters/foundation bevel), LOD2 520 (merged box+prism).

B. Town Hall — civic landmark

Size 9.0×8.0×7.5 m **Tris LOD0** 5,620 **Tex** 2048 atlas

- **Modules:** double-gable main + clock tower (cupola + bell), grand stairs, colonnade porch, banner anchors ×4, clock face (emissive at night).
- **Materials:** cream plaster, blue roof tile, stone base, gold trim.
- **LOD:** LOD1 2,600 / LOD2 900. Tower stays to last LOD (silhouette).

C. Market Stall

Size 3.5×2.8×3.0 m **Tris LOD0** 1,240 **Tex** 512

- **Modules:** 4 posts, striped canopy (fabric, double-sided), counter, goods crates (instanced), rug ground plane.
- **Canopy variants:** red/white, teal/cream, blue/cream — same mesh, 3 material IDs.
- **LOD:** LOD1 600 / LOD2 220.

D. Farm House

Size 6.0×5.0×4.5 m **Tris LOD0** 2,310 **Tex** 1024

- **Modules:** thatch roof (thick + ragged edge via alpha or geo fringe), timber frame, open veranda, barn door.
- **Materials:** thatch, rough timber, mud wall.
- **LOD:** LOD1 1,100 / LOD2 480.

E. Windmill — kinetic hero

Size 5×5×11 m + blades Ø12 m **Tris LOD0** 3,890 **Tex** 1024

- **Modules:** tapered tower (separate **BLADES** pivot object at hub), 4 lattice blades, cap roof, door/windows, stone base.
- **Rig:** blades = one object, origin at hub axle, rotate around local Z. Export pivot — Three.js spins it.
- **LOD:** blades stay; tower LOD1 1,800 / LOD2 700.

F. Lighthouse — harbor landmark

Size 4×4×12 m tower **Tris LOD0** 2,760 **Tex** 1024

- **Modules:** rock plinth (terrain kit), striped tower (white + red bands), gallery rail, lamp room (glass + emissive lamp + PointLight anchor), conical roof.
- **Light:** emissive lamp + rotating beam plane (additive, Three.js) — bake no light into geometry.
- **LOD:** LOD1 1,300 / LOD2 500.

G. Greenhouse

Size 7×4.5×3.5 m

Tris LOD0 2,180

Tex 1024

- **Modules:** timber/metal frame (instanced mullions), glass panels (single transparent material, `alpha 0.35`), interior planter rows (instanced), door.
- **Perf:** one glass draw call for whole building; no per-pane materials.
- **LOD:** LOD1 1,000 (opaque glass) / LOD2 380.

H. Storage Barn / Granary

Size 8×5×4.5 m

Tris LOD0 2,400

Tex 1024

- **Modules:** arched metal roof, timber gables, big doors, **open side platform with grain-stack anchors** (food readable from camera — never hide stock inside).
- **LOD:** LOD1 1,150 / LOD2 450.

MODELING DISCIPLINE (all buildings): bottom-anchored pivots, manifold geometry, no N-gons on deforming parts, bevel only silhouette edges, separate material IDs per region (wall/roof/timber/glass), keep UV islands on shared atlas, snow-cap child meshes named `SnowCap_*` for seasonal toggle.

4 — INFRASTRUCTURE (modular repeat kit)

Build once, instance everywhere. All timber = shared wood material. All pivots bottom-center. Snap dimensions to 1 m / 0.5 m grid.

Wooden Bridge

Module 4×3×1 m, 1,120 tris. Deck planks (instanced), 2 stringers, stone abutments, post+rail both sides. Repeat spans + end ramps. Slight arch (0.15 m camber).

Dock (T-head)

6×4×1 m, 1,480 tris. Piles (Ø0.25 m, 2 m into water), deck, cleats, lantern post, flag anchor. Pile tops above deck 0.4 m.

Pier (straight)

4×2.5×1 m, 620 tris. Same kit as dock, shorter. Chain instances for 20–30 m harbor arm.

Path Segment

2×2×0.12 m, 320 tris. Straight / curve45 / T / plaza-ring variants. Sandy top + stone edge. +0.06 m lift, real thickness.

Stairs + Platform + Railing

Stair: 2 m wide, step 0.3 tread / 0.18 riser, side stringers. Platform: 2×2 m hut decks, stilt legs. Railing: 1 m post + double rail, instanced posts.

Shoreline Structures

Breakwater rocks (terrain kit), mooring posts, rope coils (torus stack), fish-drying racks. Wet band material on all parts below +0.5 m.

Repeat Logic

Bridge = [ramp][span]×N[ramp]. Dock = [pier]×N + [T-head]. Paths snap vertex-to-vertex. Name: `ASTX_INFRA_BridgeSpan_4m` etc. One material → one draw call per type when instanced.

5 — TERRAIN (modular composition — no monolith)

Main Island Tile 16×16 m tiles, ~1,800 tris each. Grass cap + soil band + stone cliff + submerged plinth (to -5 m). Vertex-color blend grass→sand→rock. Skirt edges to hide seams.	Rocky Cliff 8×8×6 m, 1,320 tris. Angular low-poly faces, top grass lip, ledges for plants. 3 variants rotated/mirrored.	Beach Edge 10×6×2 m, 640 tris. Sand wedge overlapping grass (never floating disc), pebble strip, foam anchor edge.	Small Rocky Islet Ø 20–26 m, ~2,500 tris. Same column (plinth→cliff→cap) scaled. Hut/trees/lighthouse mount point on top.
Waterfall Cliff 6×8×10 m, 2,140 tris. Notch + lip + plunge pool (disc, foam ring), water ribbon (scrolling UV plane, separate material), mist sprite anchor.	Underwater Transition Shallow shelf to -6 m, sand→teal gradient, seabed plane. Cliffs extend below waterline — no gaps.	Reef Zone + Coral Reef Reef rocks (icosahedron deformed, 200–400 tris) + coral + seaweed instanced on shelf. Keep inside shallow ring.	Rocks / Boulders 5 sizes (0.5–3 m), 120–500 tris. Dark wet variant for waterline. Scatter + rim armor + breakwaters.

6 — VEGETATION (silhouette variation, instanced)

Do NOT make every tree identical. 3 palm lean variants, 3 broadleaf crowns, 2 bushes. Wind sway via shader (vertex color green channel = sway weight) or static + Three.js rotation for palms only.

Asset	Height	Tris LOD0	Construction
Palm Tree (×3 variants)	4–6 m	420	Curved trunk (5-seg cylinder, lean variants), 8–10 fronds (bent planes, double-sided), coconut cluster (3 spheres). Origin at base.
Tropical Broadleaf (×3)	5–7 m	680	Trunk + 3–5 blob crown spheres (displaced icospheres), 2-tone leaf material. Different crown offsets per variant.
Small Tree / Bush	2–3 m / 1–2 m	300 / 200	Bush = 2–3 blobs, no trunk. Camera-side filler, plaza exclusion respected.
Bamboo / Grass / Flowers	2–4 m / 0.4 m / 0.4 m	180 / 120 / 120	Bamboo = 5 culms + leaf cards. Grass = 3 crossed quads. Flowers = stem + 5-petal + color variants (pink/yellow/white).

Seaweed / Blue+Purple Coral / Reef Rock	0.5–1.5 m	150–350	Seaweed = wavy planes (green). Coral = branching cylinders (blue #3AA7C9 / purple #9B5DE5). Reef rock = grey-blue displaced rock + sponge dots.
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INSTANCING: palms/bushes/grass/reef = `InstancedMesh` in Three.js. One geometry + one material per species. LOD1 = billboard cross-plane at >80 m, LOD2 = hide grass/flowers.

7 — PROPS & DECOR (storytelling at 40+ px)

Rule from renders: anything that must be identified at diorama zoom must be person-sized (~2 m) or clustered. 8 px props are invisible — scale up or group.

Storage

Crate 1×1×1 m 120 tris; Barrel Ø0.8×1 m 300 tris; Sack (tapered + tie) 0.7 m; Chest (lid separate object for open anim) 350 tris. Grain-stack = cream tapered stack 2.2 m tall (NOT gold — must differ from wheat).

Market & Furniture

Table 2×1 m, Bench, Market Table with goods, Stall rug. Timber + fabric mats. Legs as separate instances.

Wayfinding

Signpost (post + 1–3 arms, painted tips), Lantern (post + glass emissive + light anchor) 60–200 tris, Flag/Banner (pole + cloth plane, wave shader), Hanging banners for Town Hall.

Harbor

Anchor (1.2 m, 250 tris), Rope coil, Buoy, Fishing rods + nets (nets = alpha plane), Treasure chest, Pots/jars, Umbrella + loungers for beach.

8 — CHARACTERS (stylized NPCs — readable from above)

No facial detail — silhouette + color role-coding only. Height ~1.7 m (slightly heroic 1.9 with hat). Rig: hip pivot, 2 legs, torso, 2 arms, head. One 1024 atlas for all NPCs. ~1,100–1,300 tris each.

NPC	Tris	Read	Accessory (separate object)
Villager Male	1,120	straw hat + blue shirt	basket
Villager Female	1,080	red top + teal skirt	pot / child hand variant
Farmer	1,240	tan hat + green shirt	hoe

Fisherman	1,180	blue cap + grey shirt	rod + creel
Guard	1,320	red helmet + spear	spear + shield

Animation (Three.js): 3 clips — idle (breath), walk (leg/arm swing), work (bend). Share skeleton across all NPCs. Export as separate GLBs per role, same rig names.

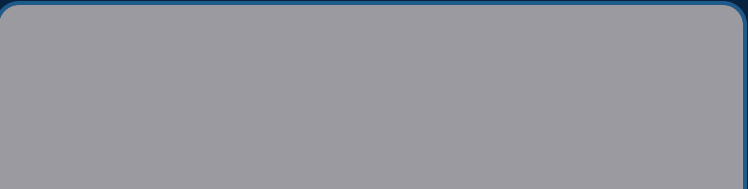
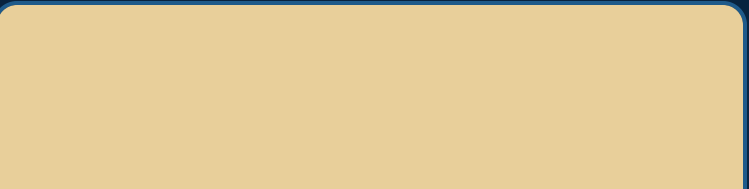
9 — BOATS

Boat	Size	Tris	Parts
Fishing Boat	5×2×2.5 m	1,160	Hull (carved box/lathe), gunwale rim, benches, outboard/cabin box, nets, lantern. Open hull shows interior.
Sailboat	6.5×3×9 m (mast)	1,420	Hull + deck + cabin, mast (separate), boom + sail (cloth, double-sided, furled morph optional), rudder, rigging lines (thin cylinders).
Rowboat	3.5×1.5×0.8 m	720	Hull, 2 benches, oars (separate for anim). Beached + moored variants.
Cargo Boat	8×3.5×3 m	1,900	Wide hull, canopy frame + fabric top, crate cargo (instanced), lanterns. Moored at dock E side.

WATER SIGNAL: at least 2 boats visible in default frame (1 moored at dock, 1 mid-channel). Scale boats UP 10–15% over true scale — at diorama zoom small boats vanish. Bob anim in Three.js (sine y + roll), not baked.

10 — MATERIAL LIBRARY (match the refs — painterly PBR)

Stylized PBR: high saturation albedo, low-frequency detail, roughness 0.6–0.9 (no mirror), no metalness except true metal. Flat + hand-painted grain via atlas, not procedural noise.

 Tropical Wood #8A5A33 · rough .8 · planks + grain streaks	 Roof Tile Orange #C96F2E · rough .75 · barrel tiles	 Roof Tile Blue #2E7FC9 · town hall / harbor	 Stone Grey #9A9AA0 · cliffs/plaza · rough .95	 Sand #E8CF9A · rough 1.0 · pebble speckle	 Water Shallow #1E9EC0 · transparent · foam edge
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Water Deep
#0A5A8A · opaque · sun glint

Grass
#5FBF5A · 2-tone tufts

Leaves Dark
#2E7D3A · crown shadow

Metal
#6B6B70 · metalness .8 · rough .4

Fabric Red
#E85D5D · canopies/flags · double-sided

Coral / Flowers
#FF6EC7 · emissive .1 · accent only

11 — COLOR PALETTE (coherent — no random hues)

Ocean deep #0A5A8A

Shallow #1E9EC0

Foam #E8FBFF

Sand #E8CF9A

Grass #5FBF5A

Foliage dark #2E7D3A

Wood #8A5A33

Timber dark #5B3418

Soil (NOT timber) #5B3418 — must differ from wood

Roof orange #C96F2E

Roof blue #2E7FC9

Stone #9A9AA0

Flower pink #FF6EC7

Harvest gold #FFB02E (only emissive accent)

Coral purple #9B5DE5

Light: warm key sun (upper-left, 48°, #FFF1D6, shadows ON) + sky fill (cool #BFE3FF, no shadow). Taxonomy: terracotta = built, green = alive, gold = harvest-ready, grey = stone, blue = water. Violet reserved for exotic/coral only.

12 — TEXTURE / UV GUIDANCE (atlas-first)

Atlas Strategy

- 1 shared **2048 atlas** for buildings (walls/roofs/timber/stone), 1×1024 for vegetation, 1×1024 for props/NPCs, 1×512 for boats. Max 4–5 textures for whole island.
- Texel density: **128 px/m** (±20%). 1 m wall = 128 px. Small props may drop to 64 px/m.
- UV: 0–1 per atlas, 4 px padding, no overlaps except mirrored/tiled parts. Hard edges → split UVs. Lightmap UV channel 2 if baked AO.

Sizes & Compression

Use	Size
Town Hall / hero	2048
Houses/farm/windmill/light	1024 (shared atlas)
Props/NPC/boats	512–1024

- Maps: Albedo (sRGB) + Normal (tangent, soft — no harsh cracks) + Roughness (packed G) + AO (baked vertex or texture, multiply). No metalness map except metal parts (uniform value).

Terrain tiles

1024 + vertex colors

Three.js: KTX2/Basis + mipmaps, anisotropy 4, sRGB albedo. One material per atlas — never per-asset duplicates.

13 — LOD STRATEGY (Three.js + glTF suitable)

Level	Screen rule	Buildings	Vegetation	Props
LOD0 full	< 40 m	Full (2-6k tris) + interior goods	Full crown + fronds	Full + goods
LOD1 mid	40-100 m (default view)	~45% tris, drop bevels/shutters, merge small parts	Simplified crown, ½ fronds	Merge to 1 mesh, drop contents
LOD2 far	> 100 m	~18% tris, box+prism, keep silhouette (tower/mast/roof)	Cross-plane billboard / blob	Hide if < 20 px, else box

Implement as `THREE.LOD` or glTF `MSFT_lod`. Share LOD1/LOD2 geometries across variants. Frustum culling per tile/asset — never merge island into one draw.

14 — BLENDER PRODUCTION NOTES (for AI modeling agent)

Naming & Collections

- `ASTX_BUILD_HouseA_LOD0`, `ASTX_VEG_Palm01`, `ASTX_PROP_Crate`, `ASTX_TER_CliffA`, `ASTX_BOAT_Sail`, `ASTX_NPC_Farmer`
- Collections: `ASTX_Buildings / Infra / Terrain / Veg / Props / NPC / Boats / Atlas`
- Materials: `M_Wood`, `M_RoofOrange`, `M_RoofBlue`, `M_Stone`, `M_Sand`, `M_Grass`, `M_Leaf`, `M_Glass`, `M_FabricRed`, `M_Metal` — share, don't duplicate.
- Modifiers kept live until export; apply on export copy.

Transforms & Cleanup

- Units meters, +Y up, +Z south (match layout). Origin = ground center (bottom-anchored). Rotation 0, Scale 1 — **Apply All Transforms** before export.
- Manifold only, no N-gons on curved parts, remove doubles, recalc normals outward, shade smooth on organic / flat on cliffs.
- Glass = separate material, `alphaMode BLEND`. Emissive lamp faces separate material `M_Emissive`.
- Windmill blades / chest lids / doors = separate objects with clean pivots, named `*_PIVOT`.

GLB Export (glTF 2.0)

File → Export → glTF (.glb): `Apply Modifiers ✓, UVs ✓, Normals ✓, Vertex Colors ✓, Materials Export ✓, Image Format WebP/JPEG (not PNG for large), Draco compression ON (buildings/terrain), LODs as separate files` `HouseA_LOD0.glb / _LOD1 / _LOD2`. Test import in three.js viewer + check draw calls < 200 total.

15 — THREE.JS REQUIREMENTS (explicit optimization)

Performance Budget

- Total draw calls < **200**, tris on screen < **800k**, textures < **30 MB** compressed.
- `InstancedMesh` for palms/bushes/grass/crates/pickets/reef. Share geometry+material.
- Frustum culling per tile; occlusion: no interiors under roofs counted as visible stock.
- Shadows: **one** directional shadow light only (2048 map, tight frustum on main island).
- KTX2 + mipmaps; LOD switching at 40/100 m; fog + distance haze for far islets.

Assembly Order in Code

- 1 seabed + water shader (depth ramp + shore foam from island footprints as uniforms)
- 2 terrain tiles → cliffs → beaches → islets → waterfalls (ribbon + mist)
- 3 paths/plaza → buildings → infra (bridges/docks) → farms → greenhouse/windmill
- 4 vegetation instancing (keep plaza/farm/harbor clear) → props → boats → NPCs → lantern/lighthouse lights
- Validate: screenshot top + persp + side; probe screen-px of key props (must be > 30 px).

ANTI-PATTERNS (will fail review): single monolithic island mesh · per-asset materials (draw-call explosion) · coplanar path decals (z-fight) · stock hidden inside roofs · 8 px props claimed visible · violet water · timber-colored soil · floating sand discs · photoreal PBR noise · facial detail on NPCs.

DELIVERABLE CHECKLIST FOR MODELING AGENT

- ☐ 8 buildings × (shaded + wireframe + dims + tris + LODs + atlas UVs)
- ☐ Terrain kit (tile/cliff/beach/islet/falls/reef/rocks) — modular, skirted
- ☐ Props (≥16) + grain-stack readable at 40 px
- ☐ 4 boats with pivots (mast/boom/oars/lids)
- ☐ Infra kit (bridge/dock/pier/path/stairs/platform/railing) with repeat rules
- ☐ Veg set (11 spp, ≥2 variants for palms/broadleaf) + sway weights
- ☐ 5 NPCs on shared rig + 3 anims
- ☐ Atlas textures (albedo/normal/rough/AO) + palette adherence

Output next: Blender .blend (collections above) + GLB set (LOD0/1/2) + atlas PNGs + this sheet as cover. Assemble in Three.js per \$15. Target: Image 3 look, Image 1+2 discipline, 60 FPS in browser.

